

BODY - SIDE VIEW

from the right · scale 0.94 : 1 · dashed = hidden · purple = mechanism

HEAD = WINDING KEY
turn it 28x clockwise
(seen from above)

NECK RATCHET
12 teeth + flexure
pawl: winds one way

RUBBER MOTOR
2 loops of #16 band
= 4 strands x 60 mm

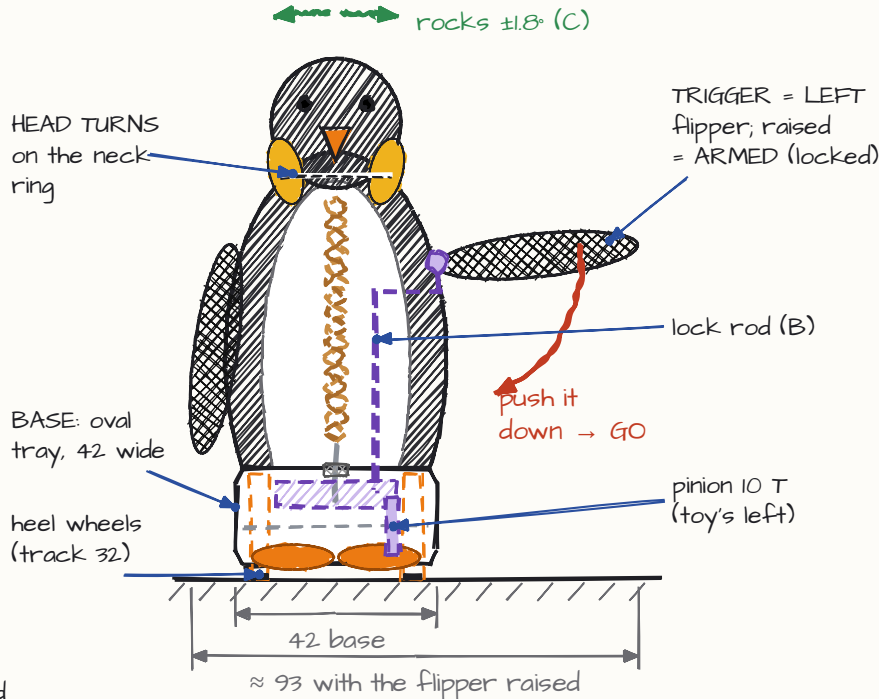
BODY: black back
shell + white belly
cap (2 snap-fit parts)

2 M8 STEEL WASHERS
low at the back
→ CG low + behind

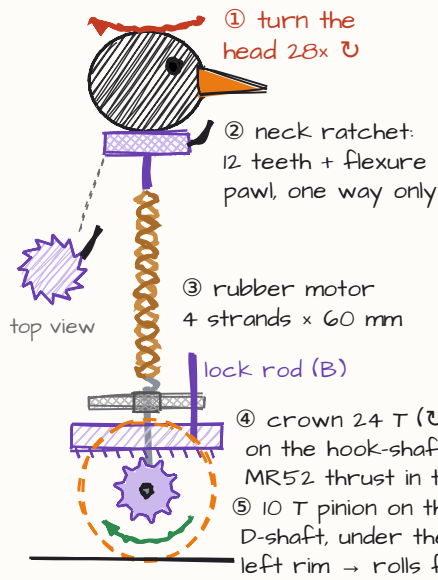
TAIL + Ø8 ROLLER
= 3rd wheel

PINION 10 T, D-bored on the heel
shaft (left side, under the crown)

FRONT VIEW (armed) scale 0.64 : 1



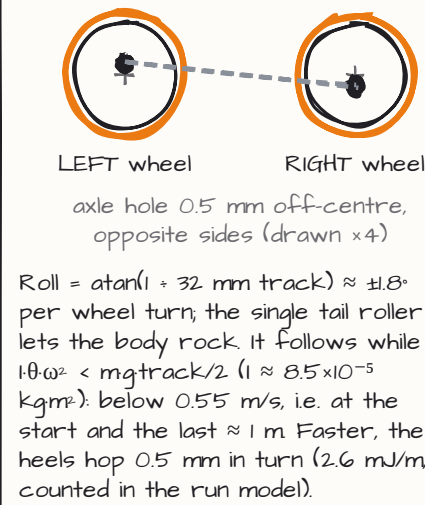
A ENERGY - TURN THE HEAD



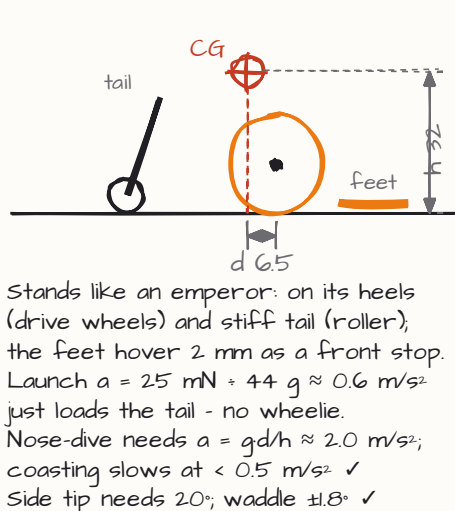
- Rubber: 2 loops of #16 band = 4 strands x 60 mm (0.34 g), hook to hook
- 28 turns (≈ 1/3 of the ~90-turn break point) → 400 J/kg x 0.34 g ≈ 135 mJ
- 400 J/kg is our low assumption; we will measure torque vs turns (a 3rd loop adds 50 % if needed)
- Drag (Crr 0.015): rolling 6.5 + heel bores 6.2 + tail roller 4.5 = 17 mN; η 0.72 incl. thrust → 2.5 m needs ≈ 62 mJ (2.2x)
- Crown 24 : pinion 10 → 2.4 wheel turns per motor turn, 28 turns = 4.6 m
- Time-step model (torque ∝ turns left, on the hook-shaft; MR52 thrust in the roof 3.1 m. Then the head free-spins.

head → ratchet → rubber → crown 24T → pinion 10T on the heel axle → wheels (x2.4)

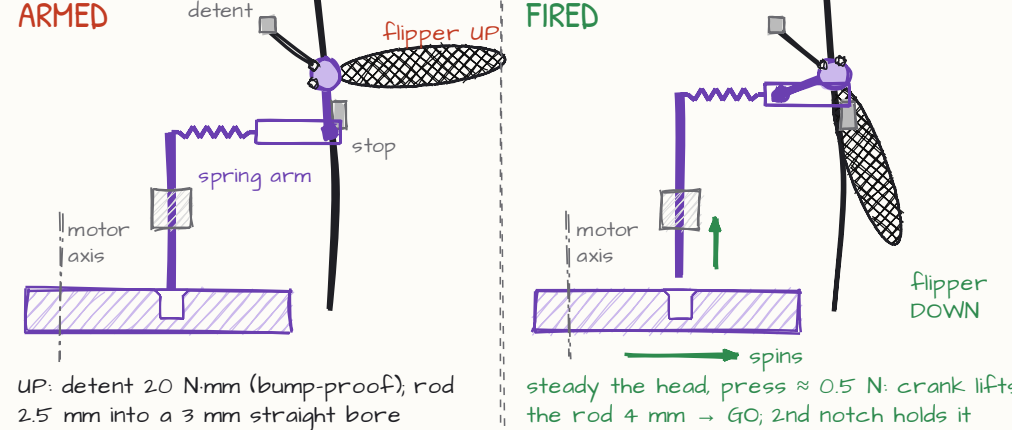
C WADDLE



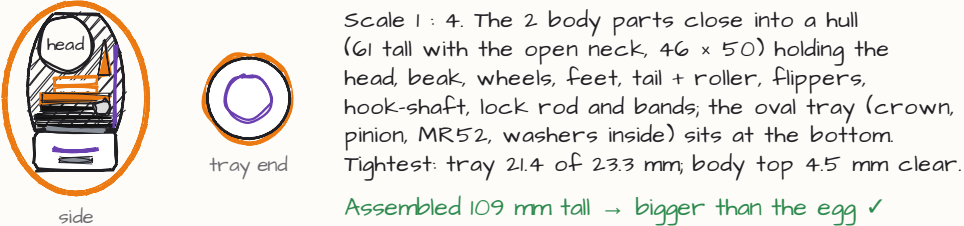
D HEELS + TAIL STANCE



B TRIGGER - FLIPPER CRANK LOCK



E PACKS INTO THE EGG



BIOMIMICRY



Emperor penguins rest on their heels and stiff tail with toes lifted, and waddle side to side. Copied: heel drive wheels + tail roller (3-point stance), low CG, rocking wheels. Bonus: the head swivels (it is the winding key).

HOW IT PLAYS

- 1 Raise the LEFT flipper - click: the lock rod springs onto the crown.
- 2 Turn the head 28x clockwise (click-click...), stop beak-forward.
- 3 Set it down - it stands on its heels and tail, fully wound.
- 4 Steady its head with a finger, press the flipper down, let go → ≈ 4.6 m.

vs Concept 1: upright body; TWISTED rubber + crown lock pin; flipper trigger.

NUMBERS

- Mass ≈ 44 g, ≈ 86 % printed PLA
- CG 32 mm high, 6.5 mm behind the heel axle (81 % on the heels)
- Motor ≈ 135 mJ; 2.5 m needs ≈ 62 mJ
- Gears ml, printed: crown 24 T → pinion 10 T (x2.4)
- Run ≈ 4.6 m (worst case 3.1 m), top speed ≈ 1.0 m/s
- Lock-rod side load ≤ 0.17 N (1.54 Nmm ÷ 9.2 mm)
- Bought (10): 3 steel shafts (heel, tail, hook), MR52, 2 O-rings, 2 bands, 2 washers

RULES CHECK

- ✓ ≥ 75 % printed PLA (≈ 86 %)
- ✓ All parts fit the egg (box E); the built toy is bigger
- ✓ Holds its energy until the flipper is pushed down
- ✓ Trigger is a flipper crank - no removable pin, stays together
- ✓ No energy stored when apart (rubber untwisted)
- ✓ Snap / press fits: no glue, no tools
- ✓ Gears are our printed parts, not commercial toy parts

ENGG*2100 F26 · Design & Build · Concept Sketch

Concept 2 · sheet 1 of 1

Concept 2: EMPEROR WADDLER

Theme: Biomimicry - emperor stance + waddle
Toy type: Running toy (≥ 2.5 m)
Scale / units: mm · side view 0.94 : 1 on 11x17

Drawn by: _____ Signature: _____ Date: _____